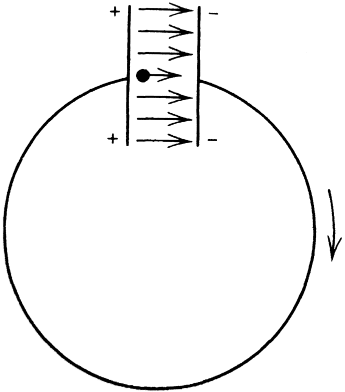
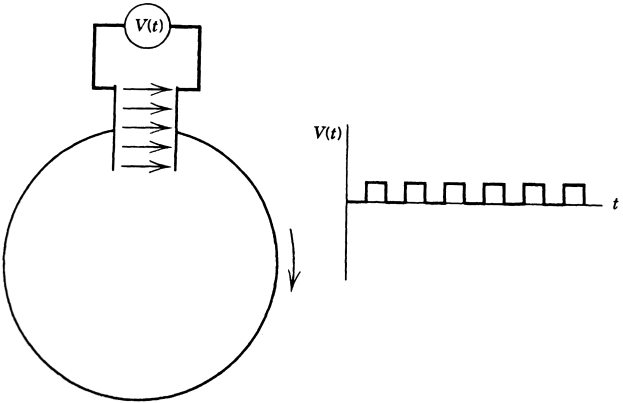
Accelerating Structures

Some initial thoughts



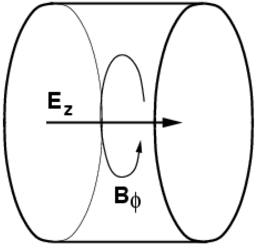
- We want nonzero field between the plates $\Rightarrow$ the plates must extend to infinity
  - No net acceleration
  - The particles are decelerated on the other side of the ring (to which the plates extend in this case)
- Finite extent $\Rightarrow$ fringe field outside the capacitor
  - Decelerates particle as it approaches and departs the capacitor
  - Still no net acceleration (as static electric field implies)
- Faraday's law: we need time-varying electromagnetic fields

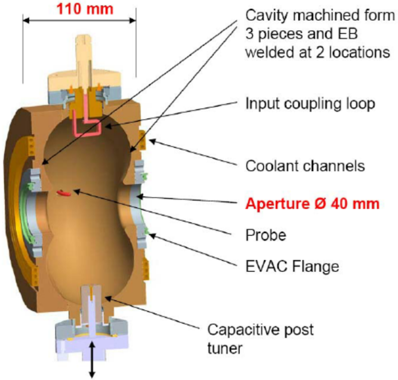
Periodic field



- What if we only turn the capacitor on when the particle is between the plates?
  - The system is inefficient, square waves are costly and non-optimal
  - No “phase focusing” for particles with slightly higher/lower momenta in the beam
  - Use resonant structures ⇒ only replace the fraction of energy extracted by the beam (predominantly)

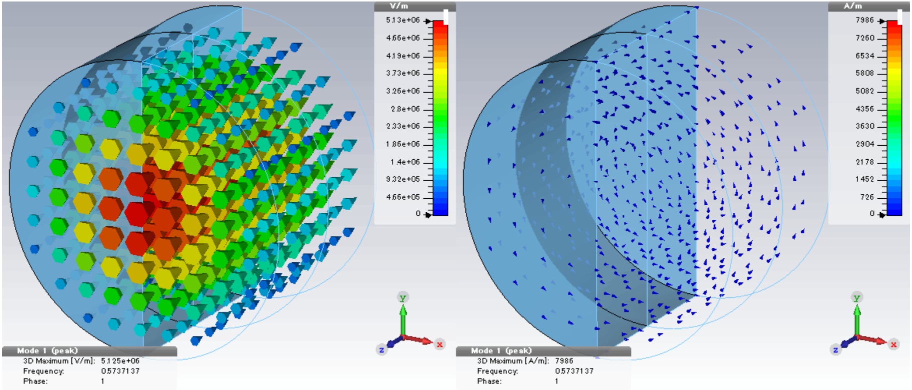
Resonant cavities



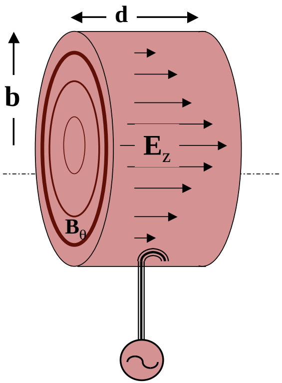


- Left: basic diagram of a pillbox cavity
- Right: a realistic cavity design (Daresbury)
- Boundary conditions: electric field is normal to the surface, zero at max radius; magnetic field is tangential to all surfaces. Can we find such field distributions?
- The next slides are borrowed from the USPAS lectures in 2009 (W. Barletta), 2011 (S. Cousineau et al.), and 2018 (M. Syphers)

Before we get into theory...



Properties of the RF pillbox cavity



$\sigma_{walls} = \infty$

- We want lowest mode: with only $\mathbf{E}_z$ & $\mathbf{B}_\theta$
- Maxwell's equations are:

$$\boxed{\frac{1}{r}\frac{\partial}{\partial r}(rB_\theta) = \frac{1}{c^2}\frac{\partial}{\partial t}E_z \quad \text{and} \quad \frac{\partial}{\partial r}E_z = \frac{\partial}{\partial t}B_\theta}$$

- Take derivatives

$$\frac{\partial}{\partial t}\left[\frac{1}{r}\frac{\partial}{\partial r}(rB_\theta)\right] = \frac{\partial}{\partial t}\left[\frac{\partial B_\theta}{\partial r} + \frac{B_\theta}{r}\right] = \frac{1}{c^2}\frac{\partial^2 E_z}{\partial t^2}$$

$$\frac{\partial}{\partial r}\frac{\partial E_z}{\partial r} = \frac{\partial}{\partial r}\frac{\partial B_\theta}{\partial t}$$

==> $$\boxed{\frac{\partial^2 E_z}{\partial r^2} + \frac{1}{r}\frac{\partial E_z}{\partial r} = \frac{1}{c^2}\frac{\partial^2 E_z}{\partial t^2}}$$

For a mode with frequency ω

- $$E_z(r,t) = E_z(r)\ e^{i\omega t}$$

- Therefore, $$E_z'' + \frac{E_z'}{r} + \left(\frac{\omega}{c}\right)^2 E_z = 0$$

  → (Bessel's equation, 0 order)

- Hence, $$\boxed{E_z(r) = E_o\ J_o\left(\frac{\omega}{c} r\right)}$$

- For conducting walls, $E_z(R) = 0$, therefore

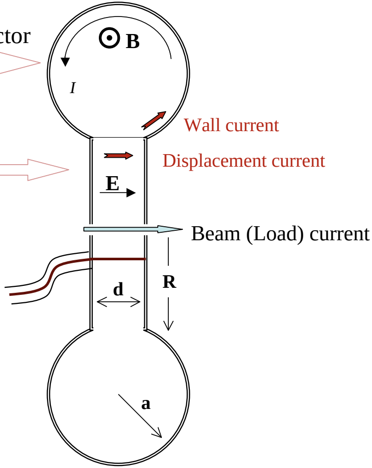
$$\frac{2\pi f}{c} b = 2.405$$

Basic components of an RF cavity

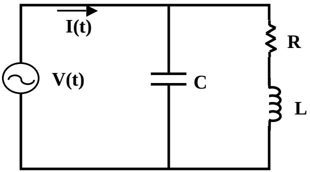
Outer region: Large, single turn Inductor

Central region: Large plate Capacitor

Power feed from rf - generator



Lumped circuit analogy of resonant cavity



$$Z(\omega) = \left[j\omega C + (j\omega L + R)^{-1} \right]^{-1}$$

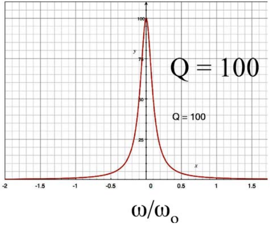
$$Z(\omega) = \frac{1}{j\omega C + (j\omega L + R)^{-1}} = \frac{(j\omega L + R)}{(j\omega L + R)\, j\omega C + 1} = \frac{(j\omega L + R)}{(1 - \omega^2 LC) + j\omega RC}$$

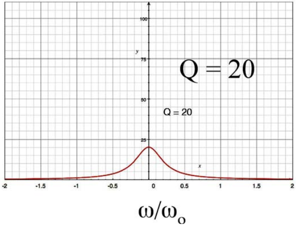
The resonant frequency is $\omega_o = 1 \big/ \sqrt{LC}$

Q of the lumped circuit analogy

Converting the denominator of Z to a real number we see that

$$\boxed{\; | Z(\omega) | \sim \left[\left(1 - \frac{\omega^2}{\omega_o^2} \right)^2 + (\omega RC)^2 \right]^{-1} \;}$$





The width is $\dfrac{\Delta\omega}{\omega_0} = \dfrac{R}{\sqrt{L/C}}$

More basics from circuits - Q

$$Q = \frac{\omega_o \circ \; Energy\ stored}{Time\ average\ power\ loss} = \frac{2\pi \circ Energy\ stored}{Energy\ per\ cycle}$$

$$\mathscr{E} = \frac{1}{2} L\, I_o I_o^* \quad \text{and} \quad \langle \mathscr{P} \rangle = \langle \mathrm{i}^2(t) \rangle R = \frac{1}{2} I_o I_o^* R_{surface}$$

$$\therefore \; Q = \frac{\sqrt{L/C}}{R} = \left(\frac{\Delta\omega}{\omega_o} \right)^{-1}$$

Translate circuit model to a cavity model: Directly driven, re-entrant RF cavity

Outer region: Large, single turn Inductor

$$L = \frac{\mu_o \pi a^2}{2\pi(R+a)}$$

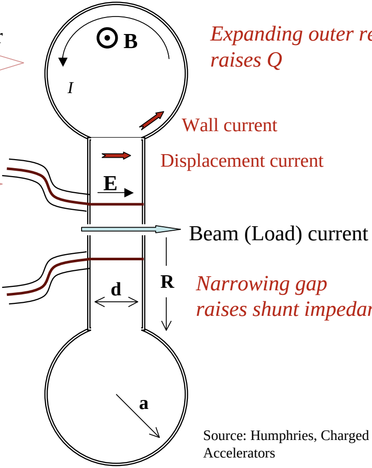
Central region: Large plate Capacitor

$$C = \varepsilon_o \frac{\pi R^2}{d}$$

$$\omega_o = 1/\sqrt{LC} = c\left[\frac{2((R+a)d}{\pi R^2 a^2}\right]^{1/2}$$

Q – set by resistance in outer region

$$Q = \sqrt{L/C}\,/R$$



Expanding outer region raises Q

Narrowing gap raises shunt impedance

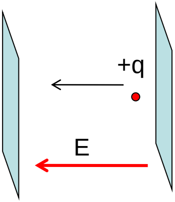
Source: Humphries, Charged Particle Accelerators

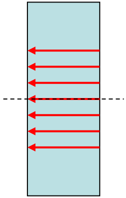
Transit Time Factor

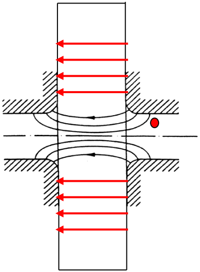
- We now consider the energy gained by a charged particle that traverses an *accelerating gap*, such as a pillbox cavity in TM_{010} mode.
- The energy-gain is complicated by the fact that the RF field is changing while the particle is in the gap.

Transit Time Factor

- We will consider this problem by considering successively more realistic (and complicated) models for the accelerating gap, where in each case the field varies sinusoidally in time.
- We also must consider the possibility that the energy gain depends on particle radius.







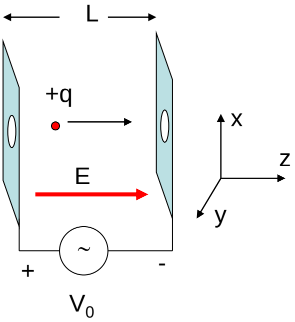
Acceleration by Time-Varying Fields

Consider infinite parallel plates separated by a distance L with sinusoidal voltage applied. Assume uniform E-field in gap (neglect holes)

$$E_z = E_z(t) = E_0 \cos(\omega t + \phi)$$

where at *t=0*, the particle is at the center of the gap *(z=0),* and the phase of the field relative to the crest is ϕ.

But *t* is a function of position *t=t(z)*, with



$$t(z) = \int_0^z \frac{dz}{v(z)}$$

The energy gain in the *accelerating gap* is

$$\Delta W = q\int_{-L/2}^{L/2} E_z dz = qE_0 \int_{-L/2}^{L/2} \cos(\omega t(z) + \phi) dz$$

Energy Gain in an Accelerating Gap

- Assume the velocity change through the gap is small, so that *t(z) = z/v,* and

$$\omega t \approx \omega \frac{z}{v} = \frac{2\pi c}{\lambda}\frac{z}{c\beta} = \frac{2\pi z}{\beta\lambda}$$

$$\Delta W = qE_0 \int_{-L/2}^{L/2} (\cos \omega t \cos \phi - \sin \omega t \sin \phi) dz$$

This is an odd-function of z

$$\Delta W = qE_0 \cos\phi \int_{-L/2}^{L/2} \cos\left(\frac{2\pi z}{\beta\lambda}\right) dz - \cancel{qE_0 \sin\phi \int_{-L/2}^{L/2} \sin\left(\frac{2\pi z}{\beta\lambda}\right) dz}$$

$$\Delta W = qE_0 \cos\phi \frac{\beta\lambda}{2\pi}\left[\sin \frac{2\pi z}{\beta\lambda}\right]_{-L/2}^{L/2}$$

Energy Gain and Transit Time Factor

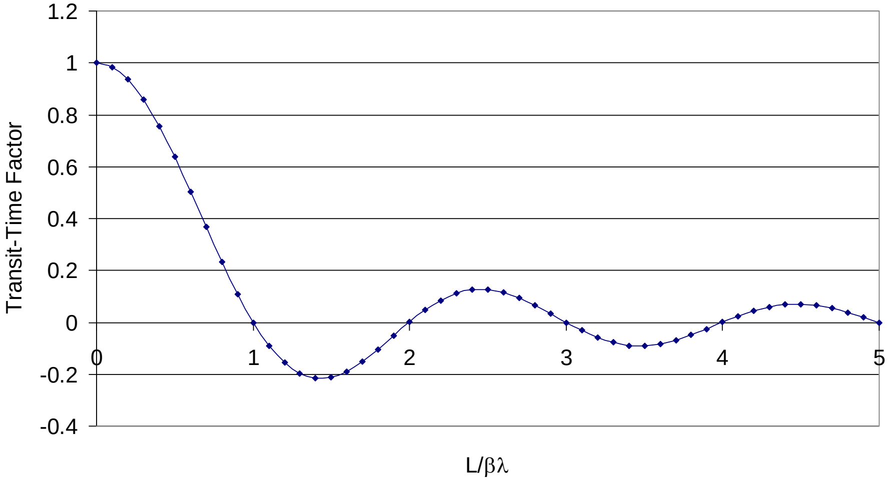
$$\Delta W = qE_0 \frac{\sin(\pi L / \beta\lambda)}{\pi L / \beta\lambda} L \cos\phi$$

$$\boxed{\begin{gathered}\Delta W = qV_0 T \cos\phi \\ T = \frac{\sin(\pi L / \beta\lambda)}{\pi L / \beta\lambda}\end{gathered}}$$

- Compare to energy gain from static DC field: $\Delta W = \Delta W_{DC} T \cos\phi$

- T is the *transit-time factor*: a factor that takes into account the time-variation of the field during particle transit through the gap.
- ϕ is the *synchronous phase*, measured from the crest.

Transit-Time Factor



For efficient acceleration by RF fields, we need to properly match the gap length *L* to the distance that the particle travels in one RF wavelength, $\beta\lambda$.

Transit Time Factor for Real RF Gaps

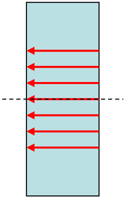
- The energy gain just calculated for infinite planes is the same as that for an on-axis particle accelerated in a pillbox cavity neglecting the beam holes.
- A more realistic accelerating field depends on *r, z*

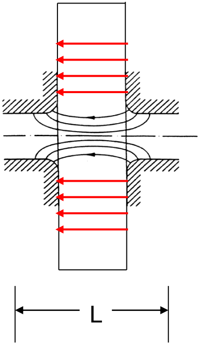
$$E_z = E_z(r,z,t) = E(r,z)\cos(\omega t + \phi)$$

- Calculate the energy gain as before:

$$\Delta W = q\int_{-L/2}^{L/2} E_z dz = q\int_{-L/2}^{L/2} E(0,z)\cos(\omega t(z) + \phi)dz$$

$$\Delta W = q\int_{-L/2}^{L/2} E(0,z)(\cos\omega t\cos\phi - \sin\omega t\sin\phi)dz$$





Transit-time Factor

- Choose the origin at the electrical center of the gap, defined as

$$\int_{-L/2}^{L/2} E(0,z)\sin\,\omega t(z)\,dz = 0$$

- This gives

$$\Delta W = qV_0 T\cos\phi = q\left[\int_{-L/2}^{L/2} E(0,z)\,dz\right]\left[\frac{\int_{-L/2}^{L/2} E(0,z)\cos\,\omega t\,dz}{\int_{-L/2}^{L/2} E(0,z)\,dz}\right]\cos\phi$$

- From which we identify the general form of the transit-time factor as

$$T = \frac{\int_{-L/2}^{L/2} E(0,z)\cos\,\omega t\,dz}{\int_{-L/2}^{L/2} E(0,z)\,dz}$$

Transit-time Factor

- Assuming that the velocity change is small in the gap, then

$$\omega t \approx \omega \frac{z}{v} = \frac{2\pi z}{\beta\lambda} = kz$$

- The transit time factor can be expressed as

$$\boxed{T(k) \equiv T(0,k) = \frac{1}{V_0}\int_{-L/2}^{L/2} E(0,z)\cos(kz)\,dz \qquad V_0 = \int_{-L/2}^{L/2} E(0,z)\,dz \qquad k = \frac{2\pi}{\beta\lambda}}$$

Radial Dependence of Transit-time Factor

- We calculated the Transit-time factor for an on-axis particle. We can extend this analysis to the transit-time factor and energy gain for off-axis particles.
- This is important because the electric-field in a pillbox cavity decreases with radius (remember TM_{010} fields):

$$T(r,k) = \frac{1}{V_0} \int_{-L/2}^{L/2} E(r,z)\cos(kz)\,dz$$

$$T(r,k) = T(k) I_0(Kr)$$

- Here I_0 is the *modified Bessel function of order zero,* and

$$K = \frac{2\pi}{\gamma\beta\lambda}$$

- Giving for the energy gain

$$\Delta W = qV_0 T(k) I_0(Kr)\cos\phi$$

which is the on-axis result modified by the r-dependent Bessel function.

Realistic Geometry of an RF Gap

- Assume accelerating field at drift-tube bore radius (r=a) is constant within the gap, and zero outside the gap within the drift tube walls

$$E(r=a,z)=\begin{cases} E_g & 0 \le |z| \le g/2 \\ 0 & g/2 \le |z| \end{cases}$$

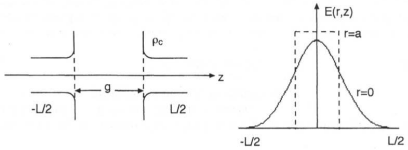


Figure 2.1 Gap geometry and field distribution.

- Using the definition of transit-time factor:

$$T(r,k)=\frac{1}{V_0}\int_{-L/2} E(r,z)\cos(kz)dz$$

- we get

$$V_0 T(k)=\frac{E_g g}{I_0(Ka)}\frac{\sin(kg/2)}{kg/2} \qquad V_0=\frac{E_g g}{J_0(2\pi a/\lambda)}$$

- Finally,

$$\boxed{T(r,k)=T(k)I_0(Kr)=I_0(Kr)\frac{J_0(2\pi a/\lambda)}{I_0(Ka)}\frac{\sin(\pi g/\beta\lambda)}{\pi g/\beta\lambda}}$$

Power and Acceleration Figures of Merit

- Quality Factor:
  - "goodness" of an oscillator

$$Q = \frac{\omega U}{P}$$

- Shunt Impedance:
  - "Ohms law" resistance

$$r_s = \frac{V_0^2}{P}$$

- Effective Shunt Impedance:
  - Impedance including TTF

$$r = \frac{(V_0 T)^2}{P} = r_s T^2$$

- Shunt Impedance per unit length:

$$Z = \frac{r_s}{L} = \frac{E_0^2}{P / L}$$

- Effective Shunt Impedance/unit length:

$$ZT^2 = \frac{r}{L} = \frac{(E_0 T)^2}{P / L}$$

- "R over Q":
  - Efficiency of acceleration per unit of stored energy

$$\frac{r}{Q} = \frac{(V_0 T)^2}{\omega U}$$

Power Balance

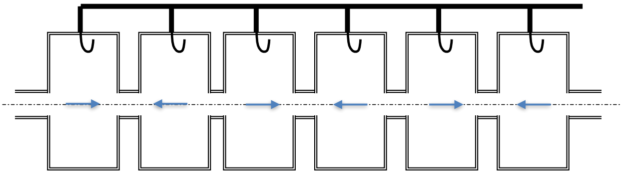
- Power delivered to the beam is:

$$P_B = \frac{I \Delta W}{q}$$

- Total power delivered by the RF power source is:

$$P_T = P + P_B$$

At high velocities, make a linac out of a series of pillboxes …



make the cells $\lambda/2$ apart

Power the cavities so that $E_z(z,t) = E_z(z)\sin(2\pi f t)$

- Once get up to higher velocities ($v \sim c$), then can consider multiple-cell cavities since the velocity is no longer changing.
- For $v = c$, the TTF will be

$$\frac{1}{\lambda/2}\int_{-\lambda/4}^{\lambda/4}\cos(2\pi z/\lambda)dz = \frac{2}{\pi}$$

Other Choices of Cavity Type

- Other variations might be required
  - particles may NOT already be at the speed of light, so transit time effects become significant
  - in a synchrotron, the speed of the particles may vary significantly during acceleration, so will require a changing RF frequency
  - slowly moving particles (like heavy ions, for instance: $v \sim 0.01c$, etc.) will need special attention
- Many variants are available; likely more to come

Superconducting Cavities

- Using superconducting materials, cavities can have Q very high ($>10^{10}$) with much lower power costs, and bunches of particles can be continuously traversing the system over time
  - "CW" -- continuous wave
- Can achieve higher acceleration per meter if use a system wherein the particle traverses several gaps, perhaps being shielded from the field while its direction changes within the cavity
- If particles were near light speed, then the cavity structure can be formed so that the gaps are separated by $\lambda/2$; elliptical cavities are common for high gradient acceleration
  - for lower speeds -- e.g., heavy ions -- must be spaced by ~ $\beta\lambda/2$. This might also limit the number of gaps per cavity to ~2

Normal vs. Superconducting Cavities

DTL tank - Fermilab

Normal conducting

Cu cavity @ 300K

$R_s \sim 10^{-3}\ \Omega$

$Q \sim 10^4$

Superconducting

Nb Cavity @ 4.2K

$R_s \sim 10^{-8}\ \Omega$

$Q \sim 10^9$

LNL PIAVE 80 MHz, β =0.047 QWR

Superconductivity allows

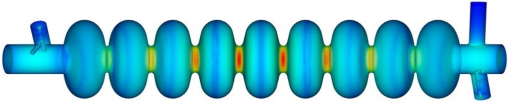
- *great reduction of rf power consumption even considering cryogenics (1W at 4.2K ~ 300W at 300K)*
- *the use of short cavities with wide velocity acceptance*

Multi-cell Cavities

- Here: "ILC" (International Linear Collider) 9-cell style cavity

as ***v —> c***, can use multiple cells in succession

cells space by RF half-wavelength



for v = c, the TTF will be… $$\frac{1}{\lambda/2}\int_{-\lambda/4}^{\lambda/4}\cos(2\pi z/\lambda)dz = \frac{2}{\pi}$$

have achieved > 35 MV/m average accelerating gradient with superconducting cavities

(Note: even larger gradients achieved with non-SC, but very power intensive)

Traveling Wave Structures

- Previous examples were standing wave structures
  - set up oscillating fields, particles arrive at pre-determined phase for optimal acceleration
- Can also feed EM energy into a structure as the beam enters; use *irises* to create phase velocity of the EM that matches the particle velocity
  - disc-loaded wave guide
    - most logical to use as particle $v \approx c$
    - *electron linacs*